

Stanley Omondi

EDUCATION

(332) 265-3572 | omondistanley.oduor@gmail.com | [linkedin](#) | [github](#)

Columbia University

B.A. in Computer Science

New York City, NY

August 2022 – May 2026

- **Coursework:** Data Structures & Algorithms, Systems Programming in C, Database Systems, Artificial Intelligence, Natural Language Processing, Deep Learning & Neural Networks, High Performance Machine Learning, Agentic AI Systems.
- **Honors:** Science Research Fellow.

RELEVANT EXPERIENCE

Sachi Health

Sept 2025 – Dec 2025

Full-Stack Software Engineer Intern

Columbia Build Lab

- Built a clinical-trial RAG pipeline that parsed research papers and normalized eligibility, symptom, treatment, and outcome data into PostgreSQL.
- Developed human-feedback review workflows for structured RAG outputs, collaborating with engineering and clinical stakeholders to refine extraction quality and HIPAA-aware recommendation requirements.
- Shipped **6 React Native and TypeScript visualization screens** for MVP testers in collaboration with product design and research teams.

Oracle

May 2025 – Aug 2025

Solutions Engineering Intern

Redwood Shores, CA

- Built AI-assisted Fast Formula workflows in Oracle Fusion HCM to generate, explain, and debug payroll logic, reducing **manual troubleshooting by 25%**.
- Implemented and regression-tested PL/SQL and HCM Fast Formulas for payroll and absence rules across test and enterprise-test environments.
- Prototyped 2 low-code AI agents and demonstrated Redwood and HCM AI capabilities to clients, translating adoption feedback into integration fixes and product notes.
- Resolved high-priority service-request defects within SLA windows, coordinating fixes and regression validation with development teams.

Columbia University

Jan 2026 – May 2026

Undergraduate Researcher(DAPLab) | Creatures in the TV

New York City, NY

- Engineered an experimental pipeline that converted single RGB images into depth-backed 2D/3D scene graphs for context-aware creature-insertion research.
- Extended a multi-model perception workflow combining GroundingDINO, SAM2, and Depth Anything V2, fusing open-vocabulary detections with per-mask depth statistics and camera back-projection into structured scene JSON.
- Prototyped depth-aware trajectory generation and isolated failure modes in occlusion, support surfaces, and motion and animation realism.
- Improved inference reliability across UDA and MPS via shared device routing, depth caching, and inter-stage model unloading, reducing VRAM pressure and runtime failures on inputs.

PROJECTS

Knowledge Distillation for Scalable Mixture-of-Expert Models

Sept 2025 – Dec 2025

[Research Paper](#)

PyTorch, Transformers, PEFT, LoRA, W&B

- Upcycled a 7B language model into a **sparse Mixture-of-Experts architecture** using pretrained expert initialization, LoRA fine-tuning, and knowledge distillation from a frozen teacher.
- Benchmarked routing, expert counts, and layer placement, achieving up to a **3.0-point MMLU improvement, 65.6% lower FLOPs, and 2.85x higher throughput**.

Pocketii – Financial Intelligence Platform

Sept 2024 – Jan 2026

[Demo](#) | [Live Platform](#)

FastAPI, Python, PostgreSQL, Plaid API, Docker, Redis, JWT

- Built a containerized finance platform with 4 FastAPI microservices behind an API gateway, ingesting bank transactions and portfolio data through Plaid and market-data integrations.
- Added Z-score and IQR anomaly detection, three-month spending forecasts, budget and low-balance alerts, and portfolio recommendations with LLM explanations and user feedback loops.

Roam – AI-Powered Social Planning App

Jan 2026 – Present

[Apple iOS TestFlight](#) | [MVP Platform](#)

Swift, FastAPI, PostgreSQL, Vision LLM, Google Maps API

- Built an iOS reel-ingestion pipeline that extracts video frames on-device with and uses a Vision LLM for confidence-scored location and activity extraction.
- Resolved locations through the Google Maps API and stored structured outputs in FastAPI and PostgreSQL to power recommendations, shared maps, and calendar-aware planning.

TECHNICAL SKILLS

Languages: Python, Bash, Java, C/C++, TypeScript, JavaScript, SQL

Machine Learning & AI: NumPy, pandas, scikit-learn, PyTorch, TorchVision, Transformers, RAG, PEFT/LoRA, vLLM

Backend, Data & Applications: FastAPI, REST APIs, PostgreSQL, MySQL, Redis, React Native, Docker, GCP

ML Systems & Developer Tools: CUDA, Linux, Git, GitHub Actions, W&B, tmux, Claude Code